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popclass: a python package for classifying microlensing events

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Summary

popclass is a python package that provides a flexible, probabilistic framework for classifying the lens of a gravitational microlensing event. Gravitational microlensing occurs when a massive foreground object (e.g., a star, white dwarf or a black hole) passes in front of and lenses the light from a distant background star. This causes an apparent brightening, and shift in position, of the background source. In most cases, characteristics of the microlensing signal do not contain enough information to definitively identity the lens type. However, different lens types can lie in different but overlapping regions of the characteristics of the microlensing signal. For example, black holes tend to be more massive than stars and therefore cause microlensing signals that are longer. Current Galactic simulations allow the prediction of where different lens types lie in the observational space and can therefore be leveraged to classify events (e.g., Lam et al., 2020).

popclass allows a user to match characteristics of a microlensing signal to a simulation of the Galaxy to calculate lens type probabilities for an event (see Figure 1). Constraints on any microlensing signal characteristics and any Galactic model can be used. popclass comes with an interface to ArviZ (Kumar et al., 2019) and pymultinest (Buchner, 2016) for microlensing signal constraints, pre-loaded Galactic models, plotting functionality, and classification uncertainty quantification methods. The probabilistic framework for popclass was developed in Perkins et al. (2024), used in Fardeen et al. (2024) and has been applied to classifying events in Kaczmarek et al. (2024).

Statement of need

The advent of the *Vera C. Rubin Observatory* (Ivezić et al., 2019) and the *Nancy Grace Roman Space Telescope* (Spergel et al., 2015) will detect tens-of-thousands of microlensing events per year (e.g., Abrams et al., 2023; Penny et al., 2019). To maximize the science output of this event stream it is critical to identify events that have a high probability of being caused by a lens type such as a black hole (Lam et al., 2022; Sahu et al., 2022), and then to allocate expensive follow-up observations such as space-based astrometry (e.g., Sahu et al., 2022) or ground-based adaptive optics imaging (e.g., Terry et al., 2022) to confirm their nature.

Current microlensing software packages such as DarkLensCode (Howil et al., 2024) or PyLiMASS (Bachelet et al., 2024) estimate lens mass-distance constraints using microlensing event light curve and additional auxiliary information (e.g., source proper motions, distances, color, or finite source effects). Using auxiliary information makes these current methods powerful but

limits them to only be effective for events with the available auxiliary data. Moreover, no current software tools explicitly predict lens type, and they always assume a fixed Galactic model. `popclass` fills the need for a flexible microlensing classification software package that can be broadly applied to classify all events from the Vera C. Rubin Observatory and *Nancy Grace Roman Space Telescope* and can be used with any Galactic model in the form of a simulation.

Method

`popclass` relies on the Bayesian classification framework in (Perkins et al., 2024). Consider the data from a single microlensing event d, and a model of the Galaxy $\mathcal{G}$. `popclass` calculates the probability that the lens of the events belongs to each lens class, class_L, where $\text{class}_L \in \text{classes}$ and, for example, $\text{classes} = \{\text{Star, Neutron Star, White Dwarf, Black Hole}\}$. Namely, `popclass` calculates

$$p(\text{class}_L|d, \mathcal{G}) \text{ for } \text{class}_L \in \text{classes}.$$

Using Bayes' theorem,

$$p(\text{class}_L|d, \mathcal{G}) = \frac{p(\text{class}_L|\mathcal{G})p(d|\text{class}_L, \mathcal{G})}{p(d|\mathcal{G})}.$$

Assuming that the set of considered lens classes is complete, $p(d|\mathcal{G})$ is a normalization factor such that all lens class probabilities sum to unity. Using importance sampling (e.g., Hogg et al., 2010) with S independent posterior samples $\theta_c \sim p(\theta|d)$ drawn under some prior, $\pi(\theta)$, obtained from fitting some set of microlensing signal parameters, θ,

$$p(\text{class}_L|d, \mathcal{G}) = \frac{p(\text{class}_L|\mathcal{G})}{p(d|\mathcal{G})} \times \frac{1}{S}\sum_{c=0}^{S} \frac{p(\theta_c|\text{class}_L, \mathcal{G})}{\pi(\theta_c)}$$

.

This allows the use of previously calculated posterior samples to perform lens classification for a single event in the context of a Galactic model. The term $p(\theta_c|\text{class}_L, \mathcal{G})$ is calculated using kernel density estimation in `popclass` over θ with a simulated catalog of microlensing events from $\mathcal{G}$. $p(\text{class}_L|\mathcal{G})$ is the prior probability that a event belongs to each class before any data is seen, which is just set by relative number of expected events predicted by the Galactic model $\mathcal{G}$.

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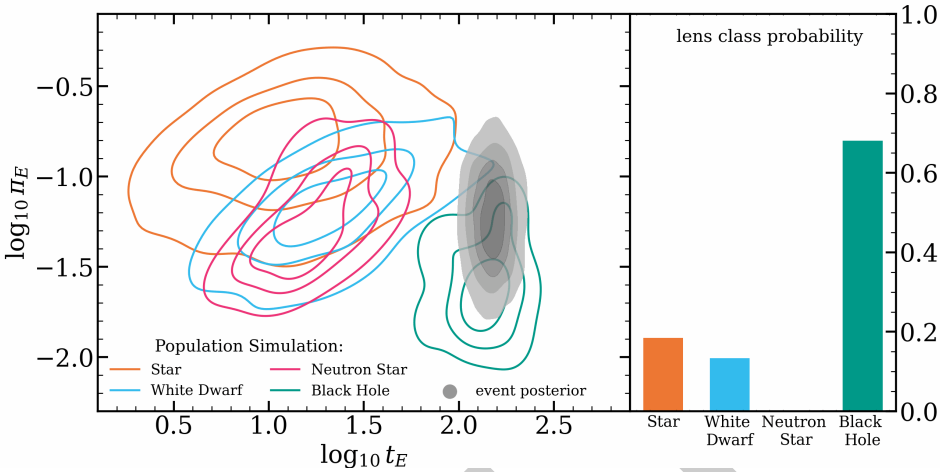

Figure 1: Left: posterior distribution of an event in log10(timescale)-log10(parallax) space, overlaid on 'star', 'white dwarf', 'neutron star' and 'black hole' contours. Right: bars showing probabilities of that event belonging to each of the lens populations.

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